Thick lines: $m_1 = 10^{-6}$ eV; thin lines: $m_1 = 0.03$ eV

Relics at rest — Relics in motion, helicity ignored — Relics in motion, helicity kept

Dirac scalar, $M = 31.6$ keV, $g = 2 \times 10^{-3}$

Dirac scalar, $M = 4.22$ MeV, $g = 1.35 \times 10^{-3}$

Dirac vector, $M = 31.6$ keV, $g = 2 \times 10^{-3}$

Dirac vector, $M = 4.22$ MeV, $g = 1.35 \times 10^{-3}$

Flux / flux without interaction

1.0
0.5
0.0

13
14
15
16

Neutrino energy, $\log_{10}(E/\text{eV})$